

CALIFORNIA REAL ESTATE **PRICE PREDICTION**

IDX Exchange DS 31

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PROBLEM STATEMENT

Goal

Build a model to predict a property's Close Price based on its physical and categorical characteristics

Context

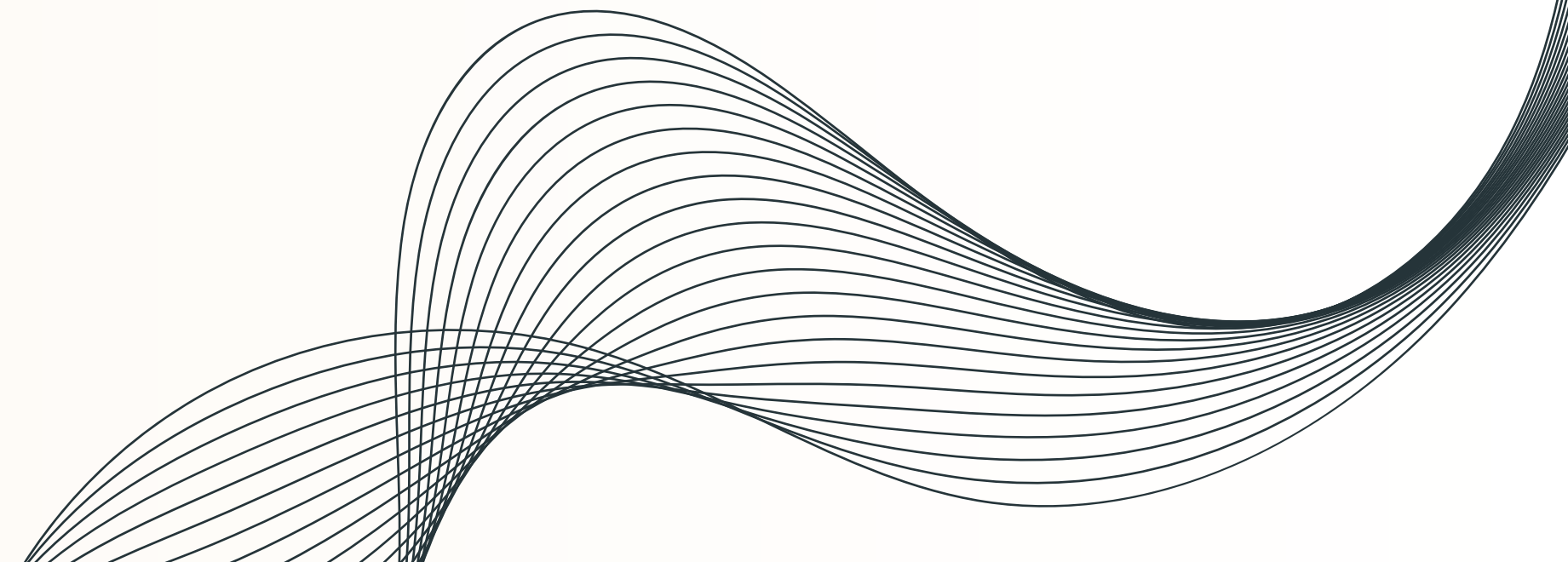
This analysis focuses exclusively on California properties classified as Single-Family Residences

Importance

Helps IDXExchange evaluate property investments, produce price estimates, and reduce guesswork

DATA OVERVIEW

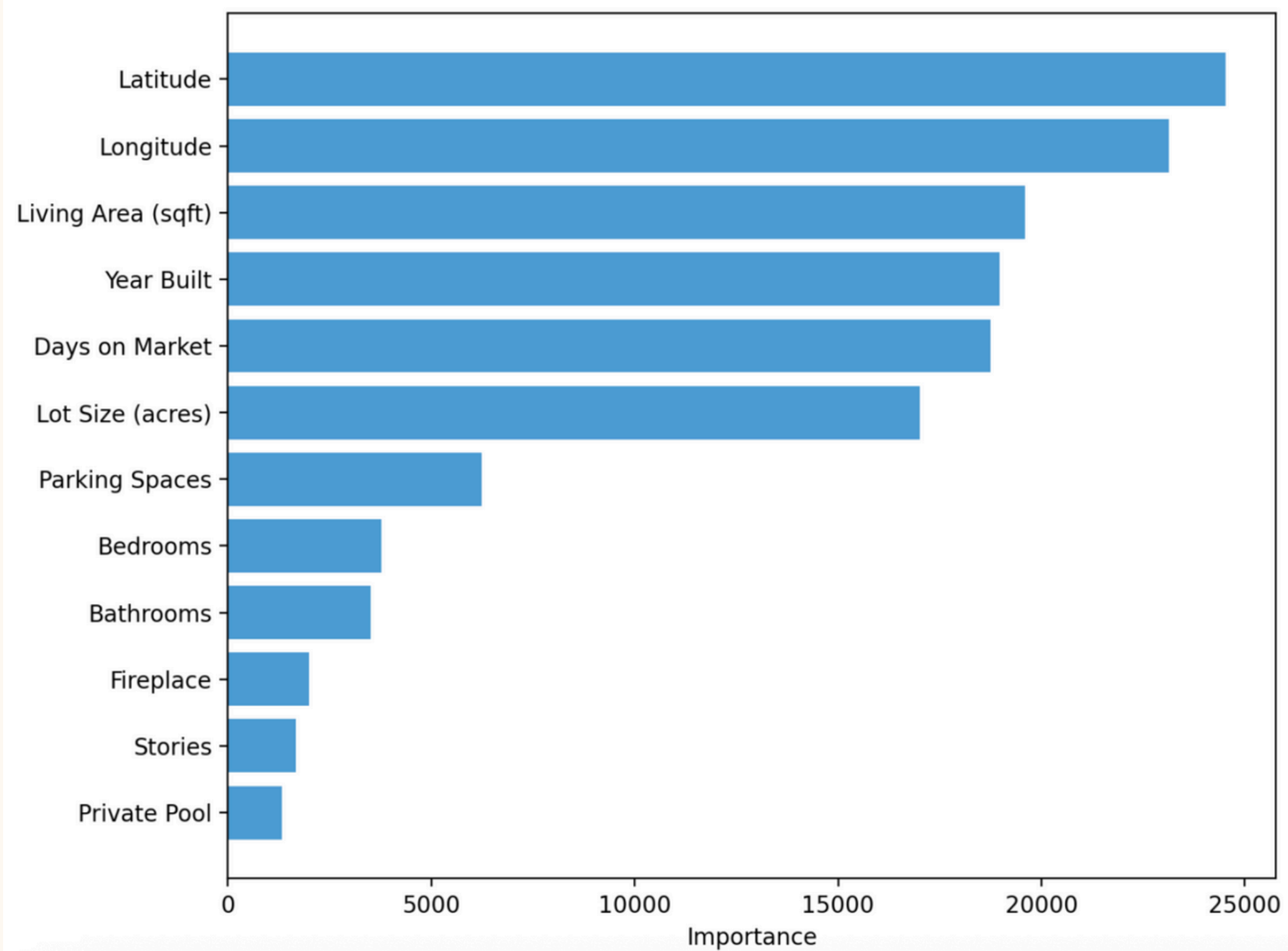
- **Data source:** CRMLS (California Regional Multiple Listing Service)
- **Time period:** Jan. - Oct. 2025
- **Geography:** California
- **Property type:** Residential - Single Family Residence
- **Sample size:**
 - Training dataset → Jan. - Aug. (~86,000 observations)
 - Testing dataset → Sept. - Oct. (~23,000 observations)



VARIABLE SELECTION

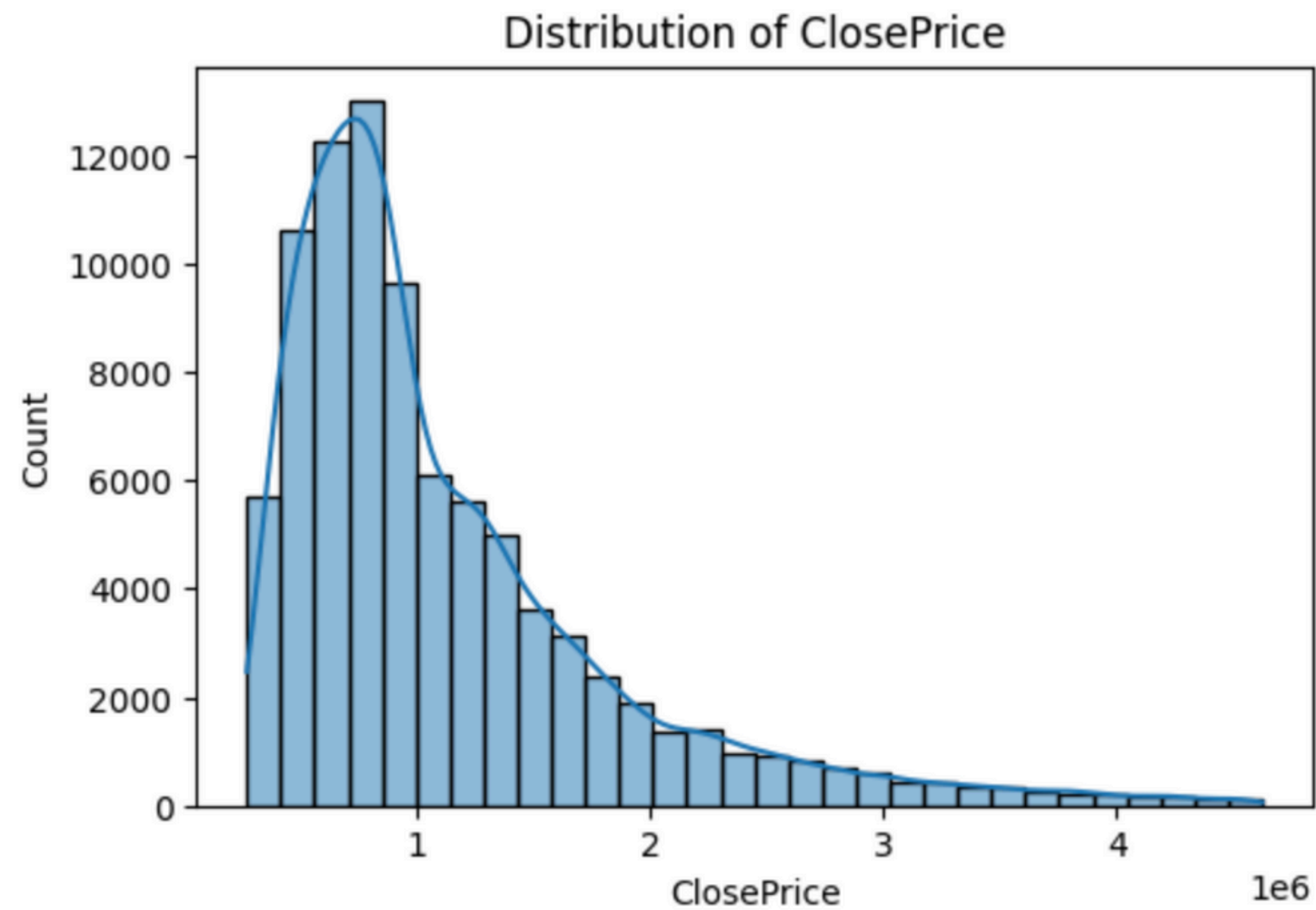
- Target: Close Price
- Main features:
 - Latitude: The geographic latitude, specified in degrees and decimal parts
 - Longitude: The geographic longitude
 - Living Area: The total livable area within the structure
 - LotSizeAcres: The total Acres of the lot
 - Stories: The number of floors in the property being sold
 - BedroomTotal: The total number of bedrooms in the dwelling
 - BathroomsTotal: The simple sum of the number of bathrooms
 - Fireplace: Does the property include a fireplace
 - PoolPrivate: The property has a privately pool that is included in sale
 - ParkingTotal: The total number of parking spaces included in the sale
 - DaysOnMarket: The number of days the listing is on market
 - YearBuilt: The year that an occupancy permit is first granted for the house

FEATURE IMPORTANCE



EDA

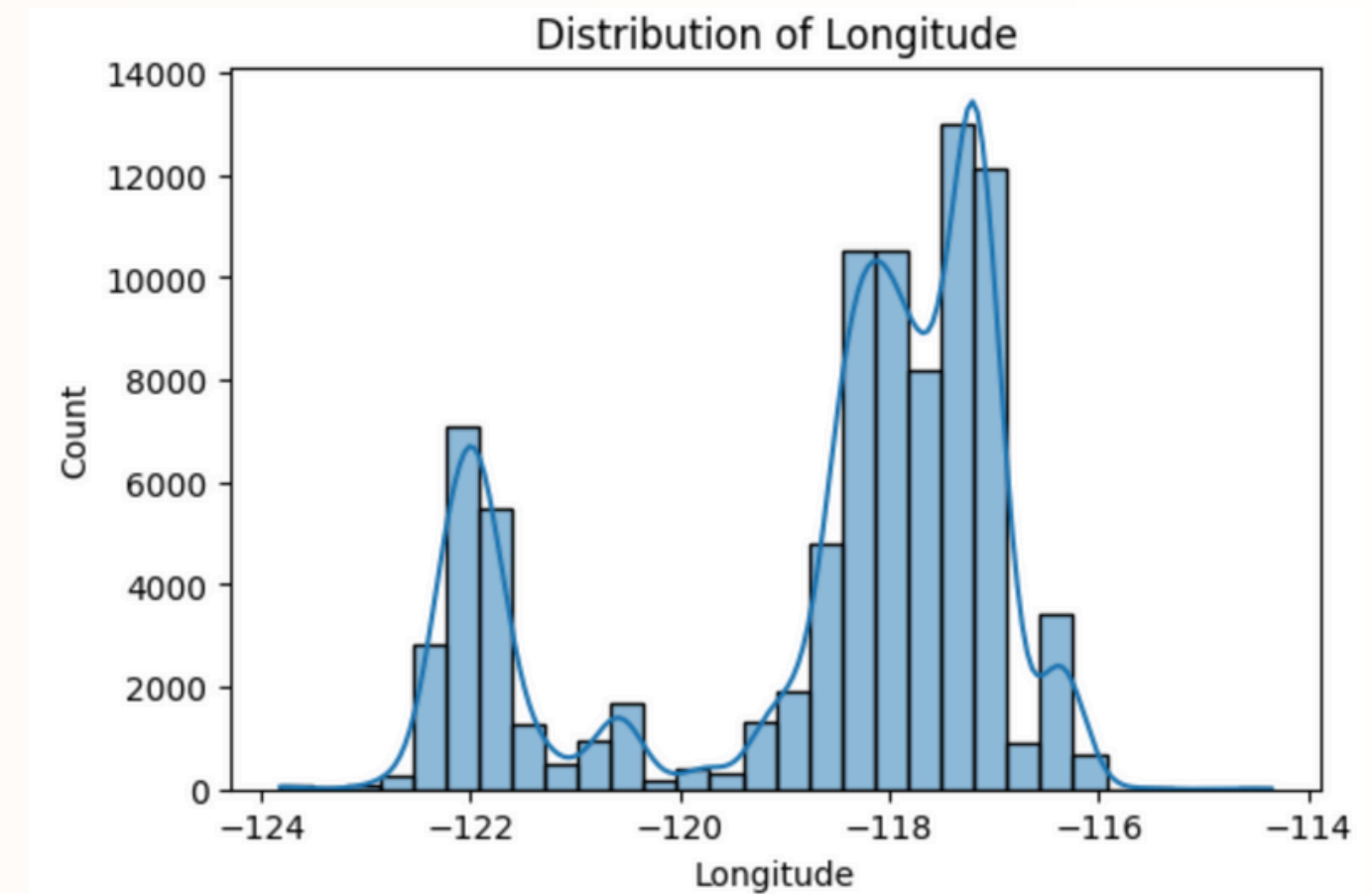
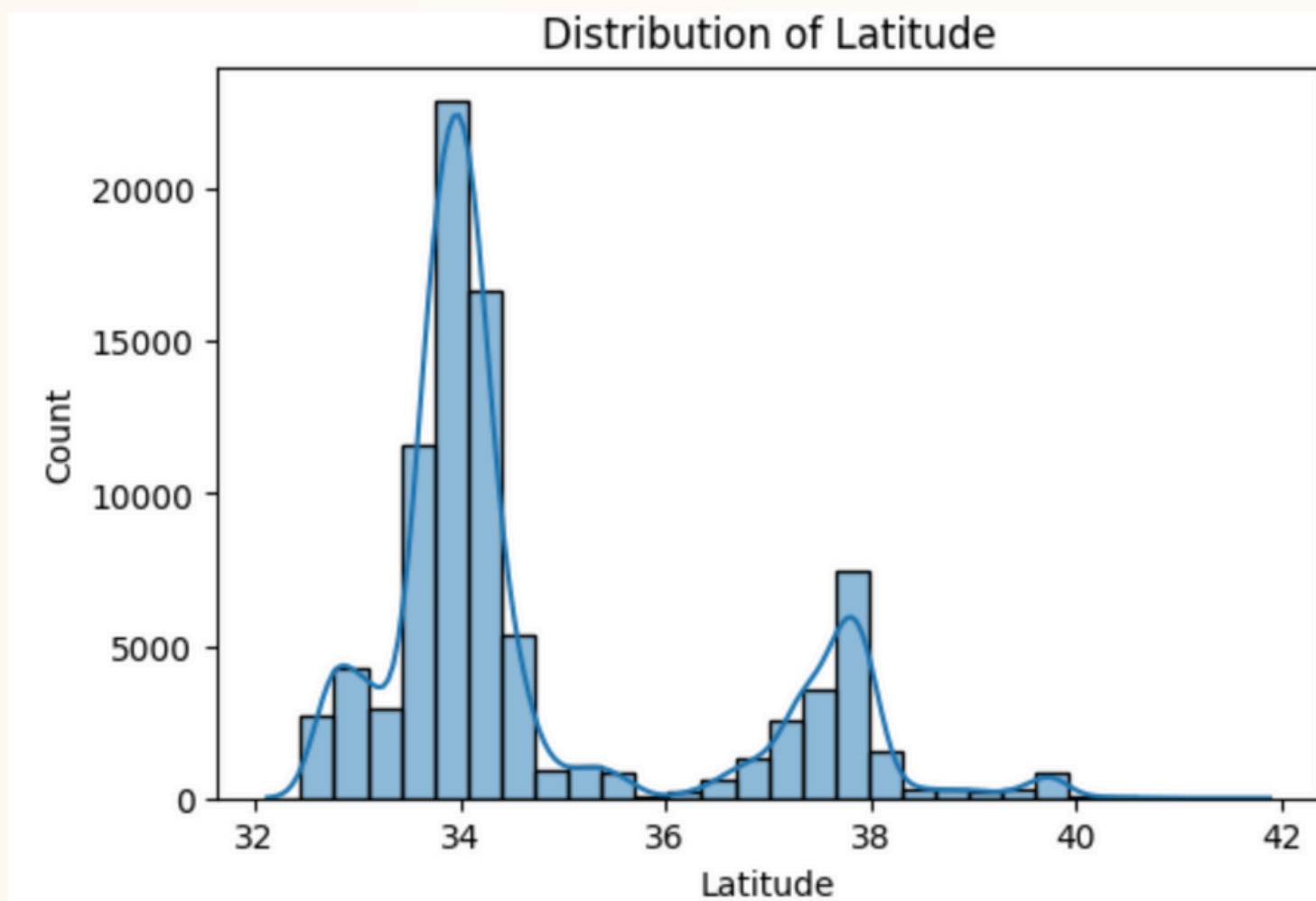
Target: Close Price



- Right skewed
- Need transformation

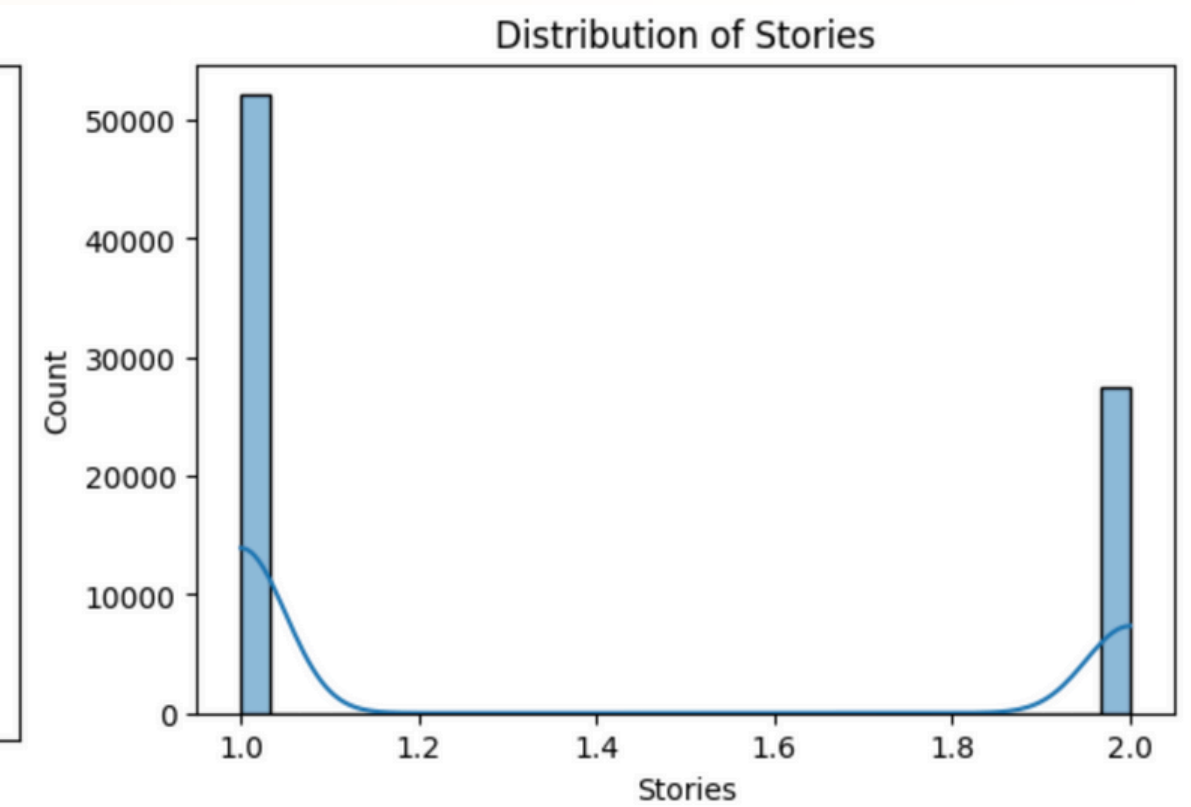
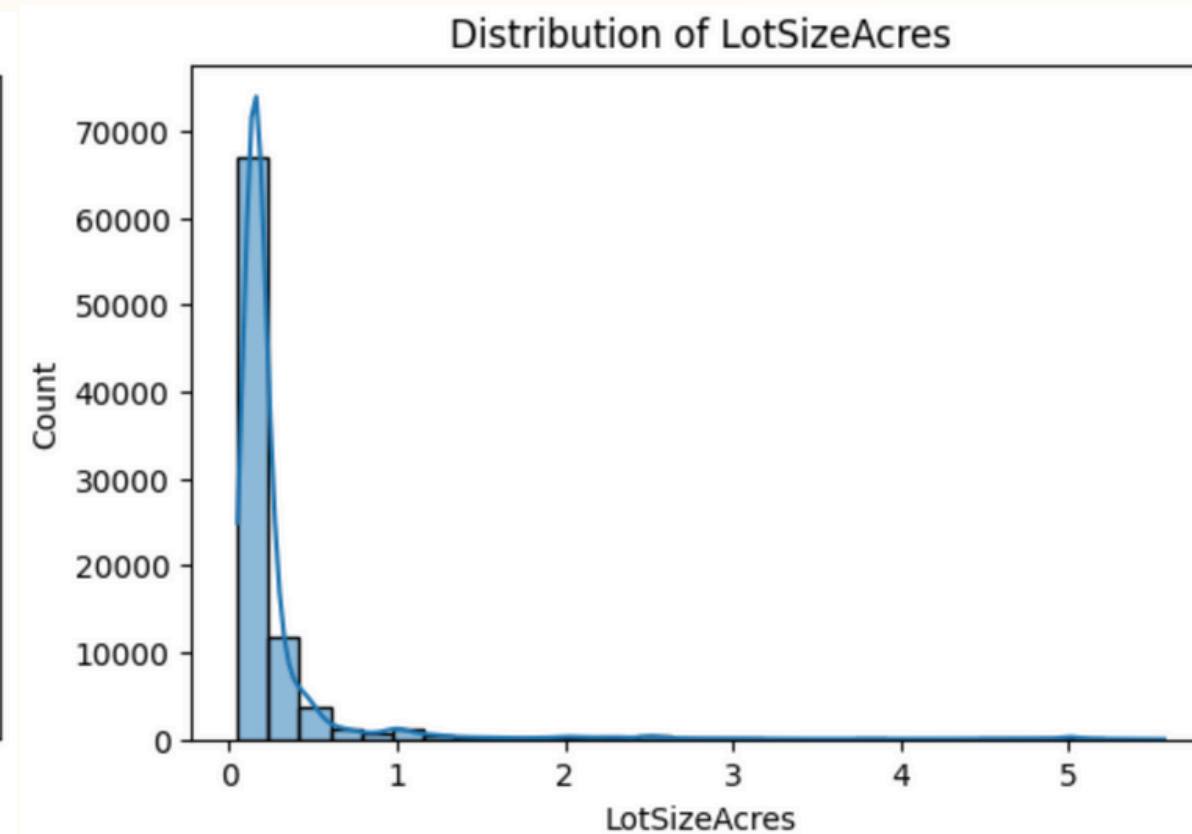
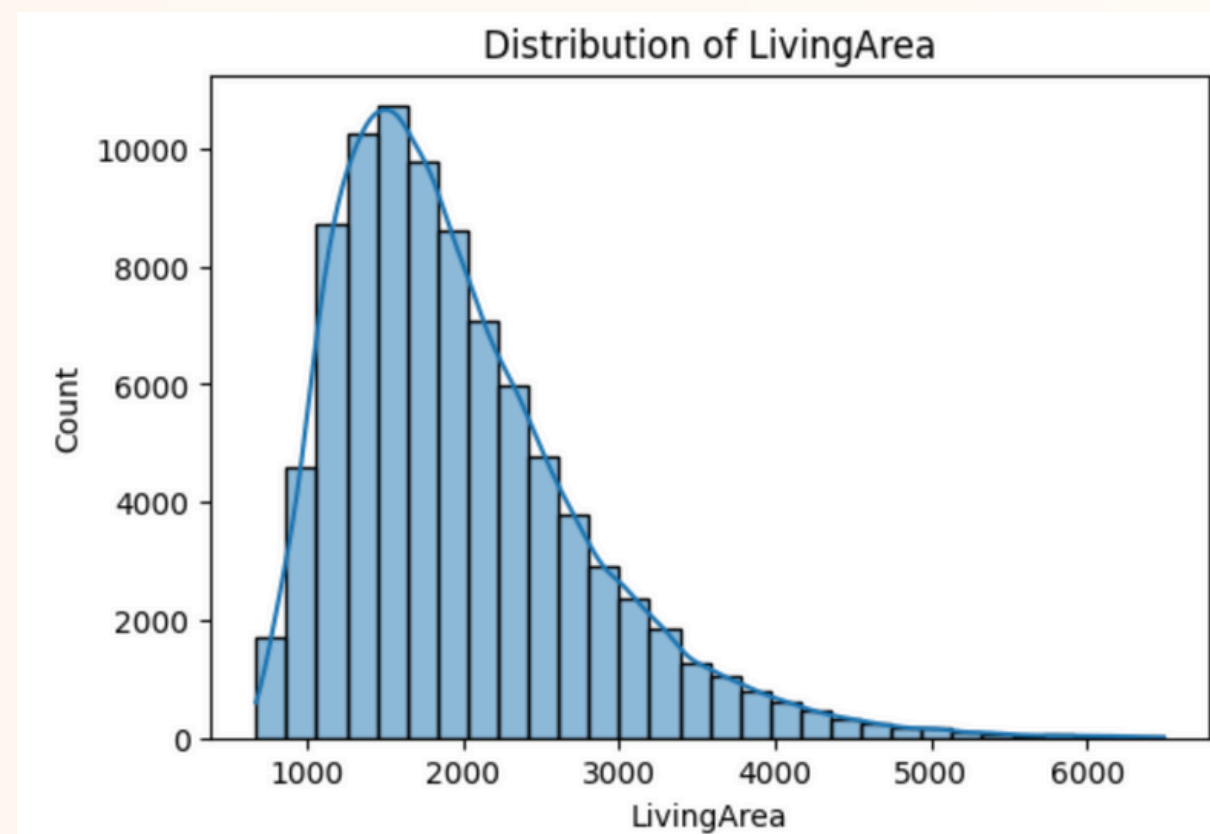
EDA

Location predictors: Latitude, Longitude



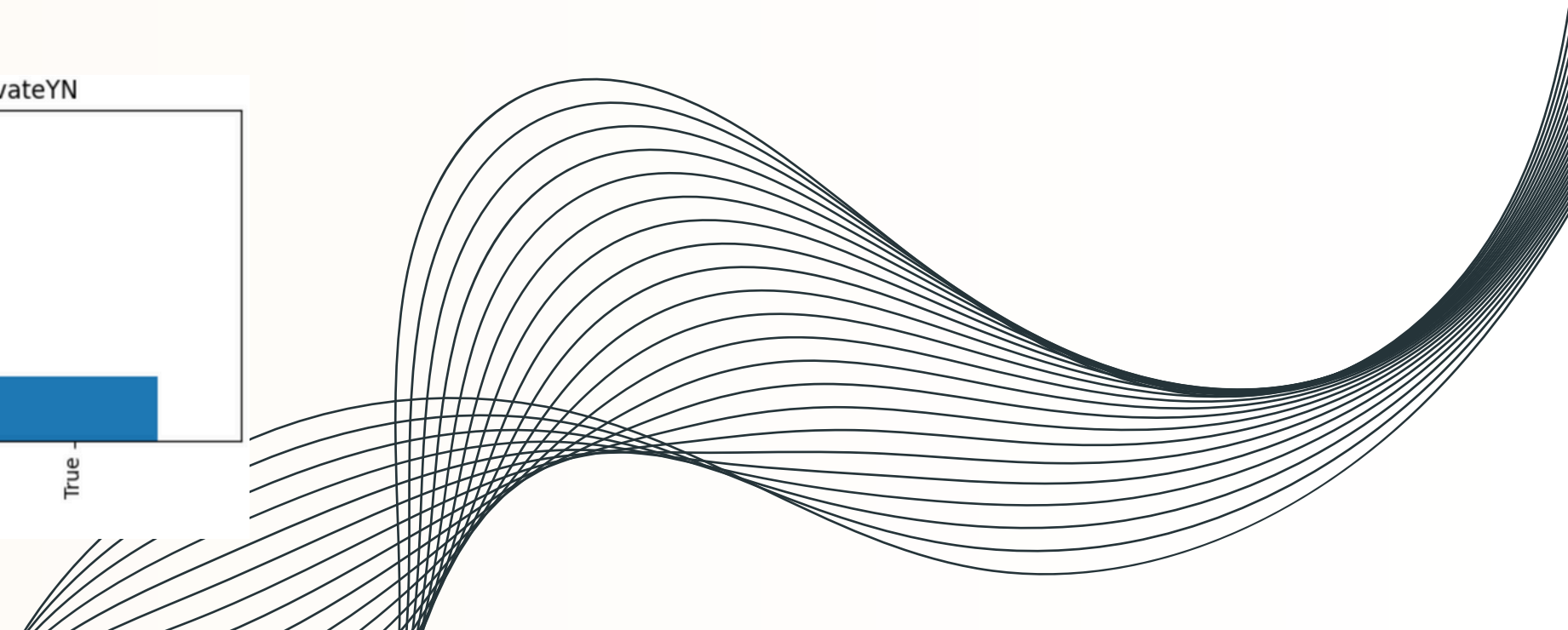
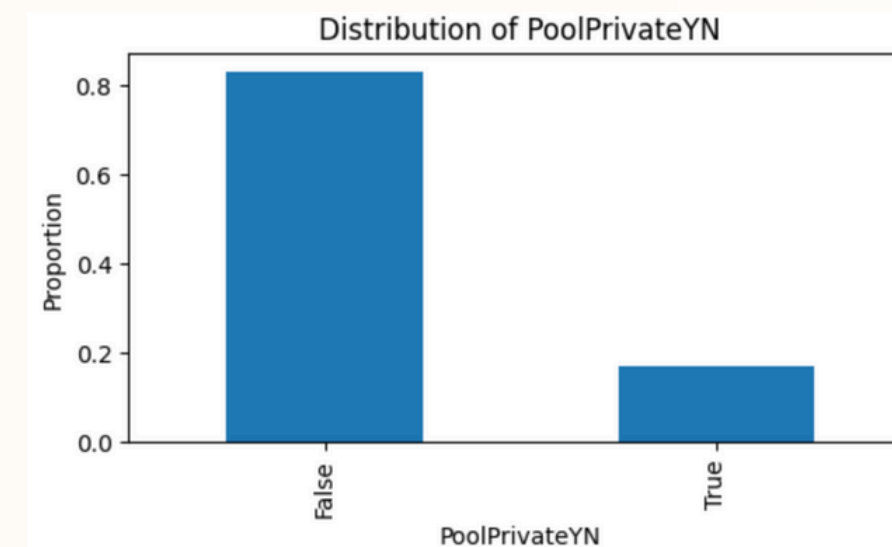
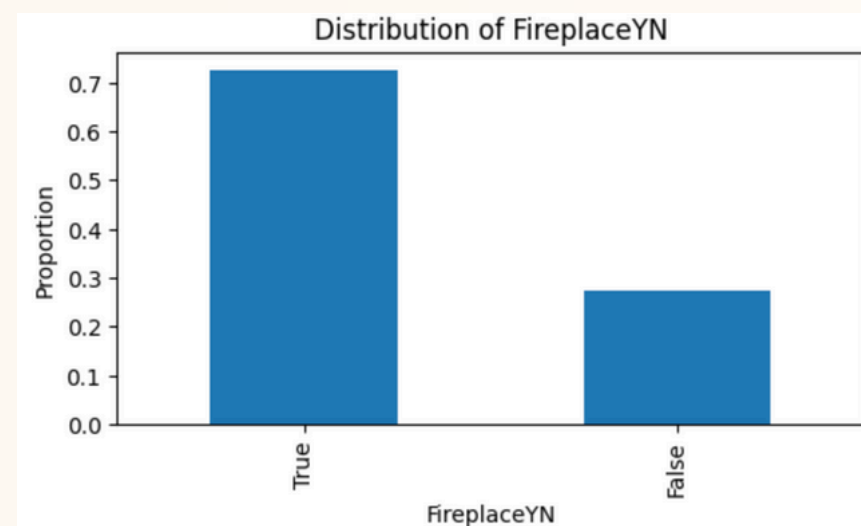
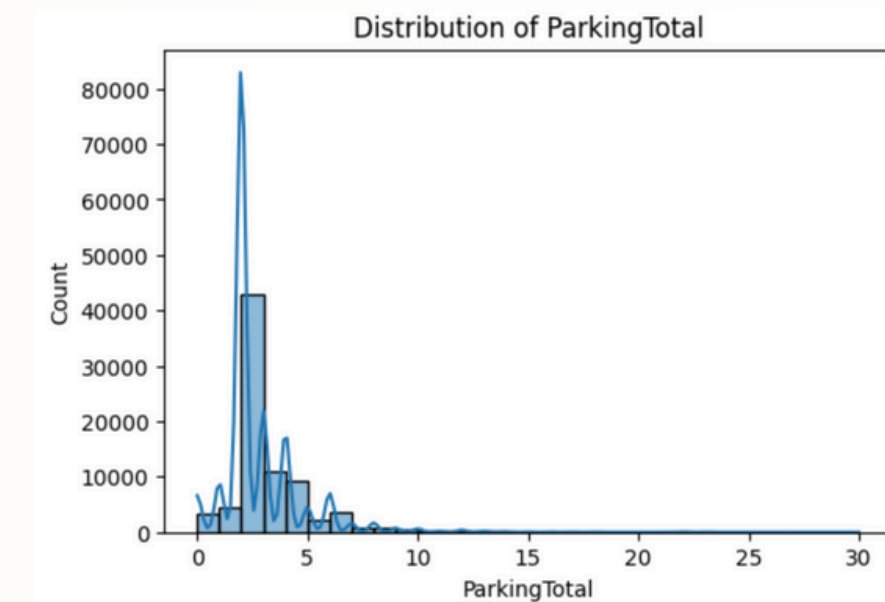
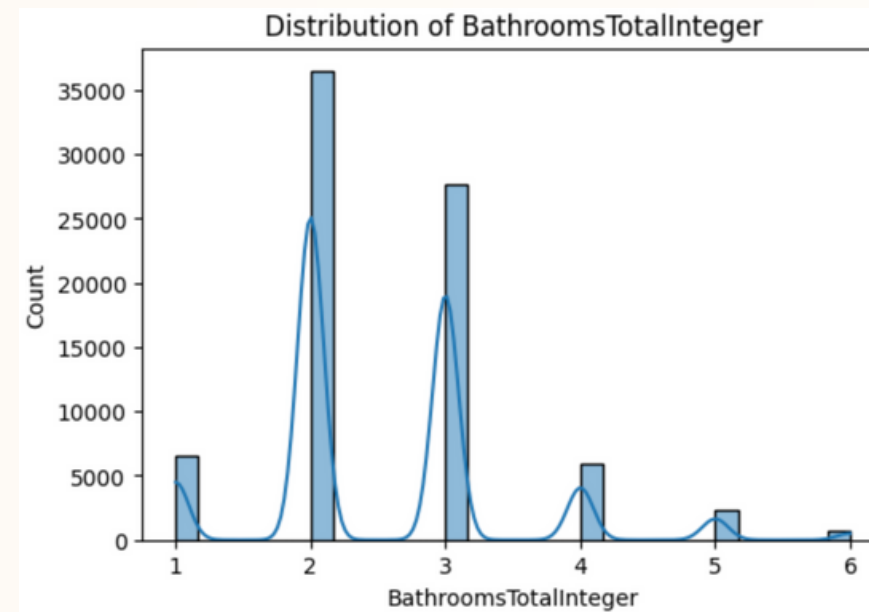
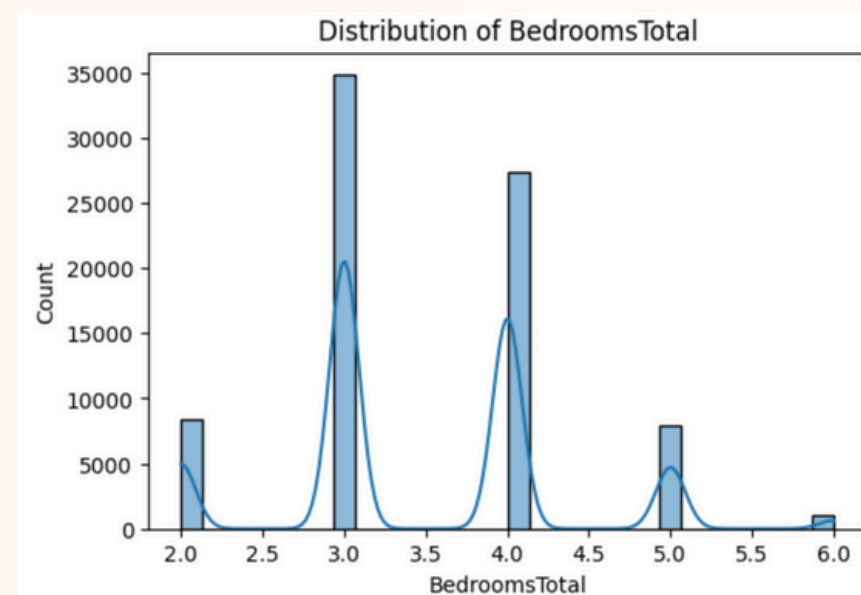
EDA

Size predictors: LivingArea, LotSizeAcres, Stories



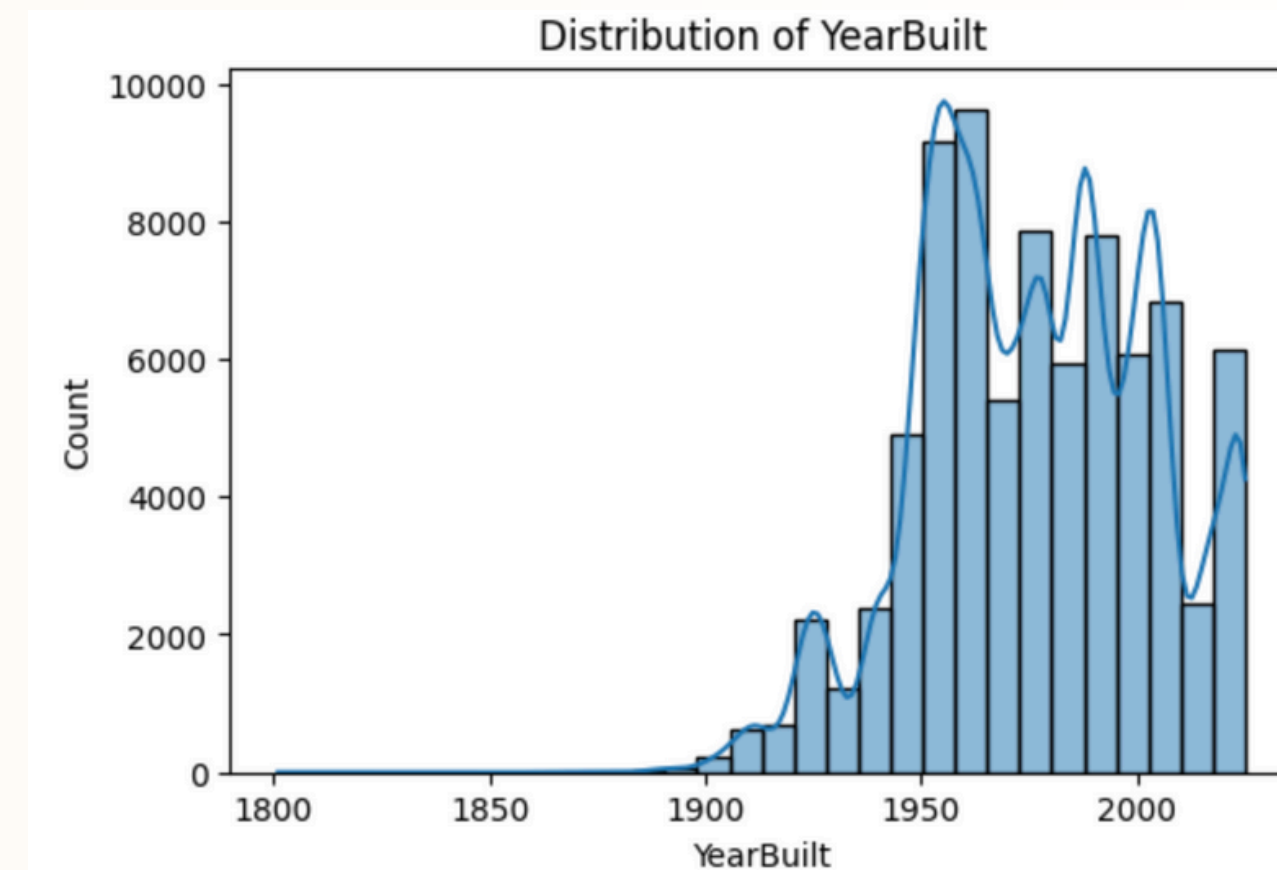
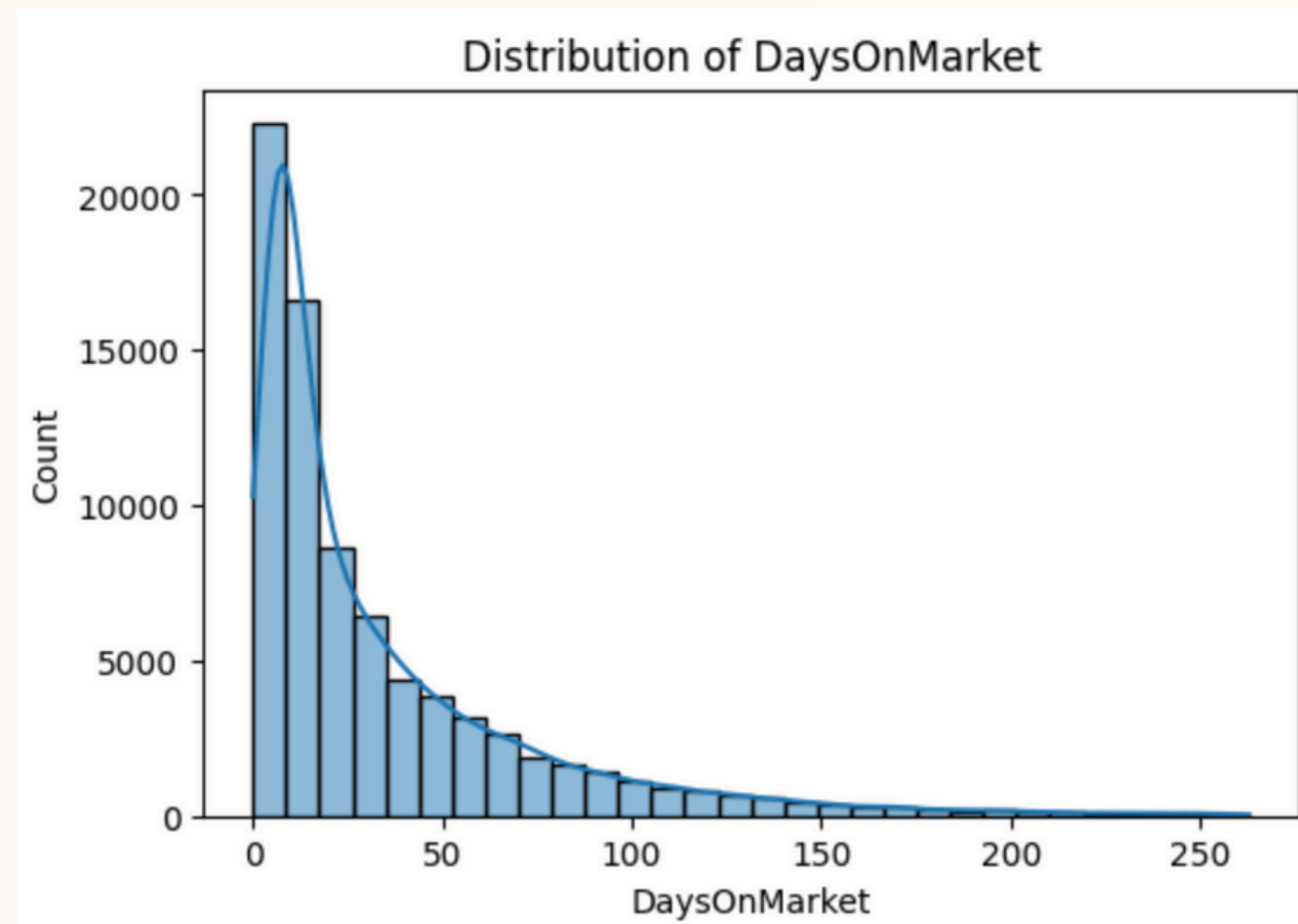
EDA

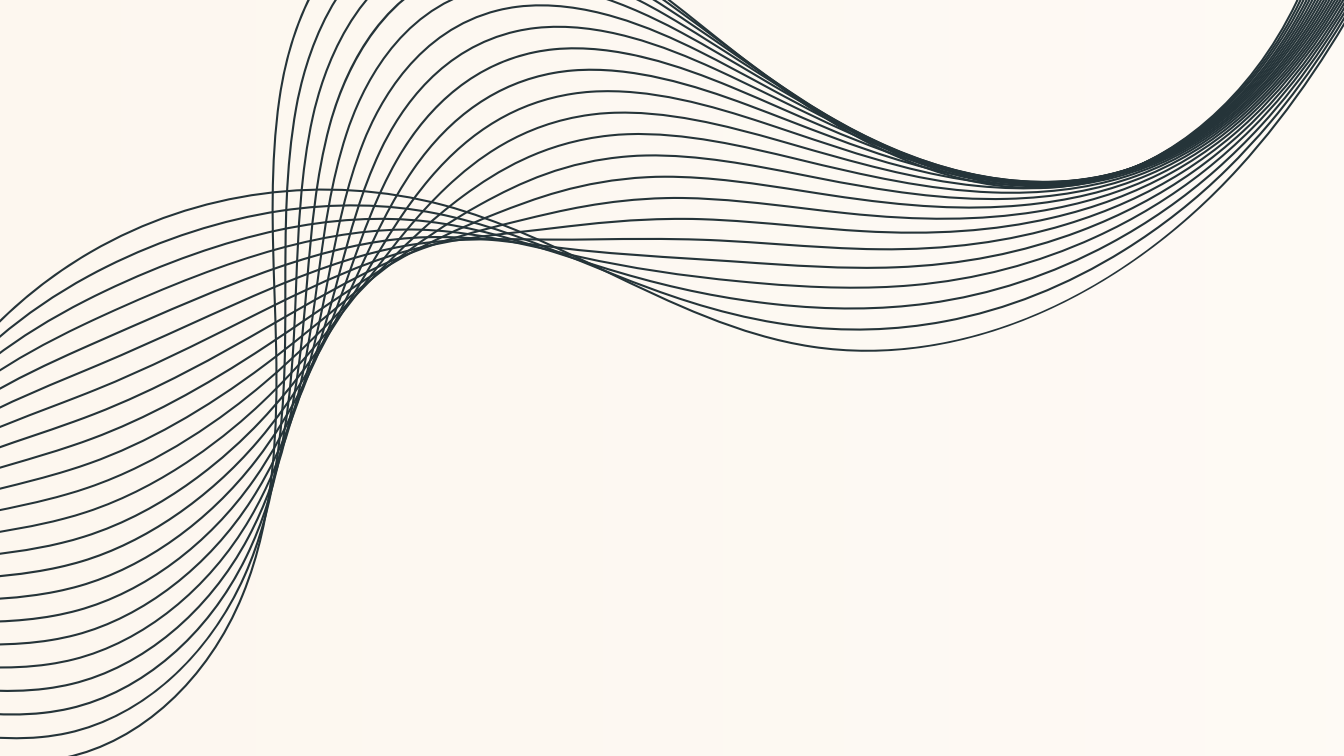
Structure predictors: BedroomsTotal, BathroomsTotalInteger, FireplaceYN, PoolPrivateYN, ParkingTotal



EDA

Market predictors: DaysOnMarket, YearBuilt





BASELINE PERFORMANCE: LINEAR REGRESSION

- We used linear regression as a benchmark model for comparison
- Assumes a linear relationship between features and home price
- Showed strong underfitting on housing data
- Revealed need for nonlinear models
- Motivated our shift to ensemble models



BASELINE PERFORMANCE: LINEAR REGRESSION

Trimmed Data Evaluation (Middle 99%)

- Overall Performance
 - R-squared: 0.403
 - MAPE: 30.94%
 - MdAPE: 23.60%

Price Bands	MAPE(%)
<\$500K	53.11
\$500K-\$1M	17.42
\$1M-\$2M	23.74
\$2M-\$5M	42.7

Key Takeaway: Performance is strongest in the mid-price ranges (\$500K-\$2M) and weak at the low and high extremes (even after trimming)

MOVING FORWARDS...

XGBoost

- **Sequential Learning:** Unlike Random Forest, XGBoost builds trees one at a time, where each new tree specifically corrects the errors (residuals) of the previous one
- **Built-in Regularization:** Includes L1 (Lasso) and L2 (Ridge) regularization to prevent overfitting, making it more robust against noise in the housing data
- **Why We Chose it:** It balances model complexity with generalization, allowing it to capture the specific price premiums of unique properties better than the baseline

LightGBM

- **Leaf-Wise Growth Strategy:** While other models grow trees level-by-level, LightGBM grows "leaf-wise", which focuses on the branch with the highest error first and reduce error faster
- **Histogram-Based Algorithms:** Buckets continuous feature values into discrete bins, reducing memory usage and training speed
- **Scalability:** significantly faster training times than XGBoost, allowing for more rapid hyperparameter tuning
- **Why We Chose It:** Its aggressive growth strategy often yields lower MAPE and MdAPE

FINAL MODEL: XGBoost

1

MODEL CONFIGURATION

- **Algorithm**
 - XGBoost
- **Split**
 - Jan-Aug (Train)
 - Sep-Oct (Test)
- **Target**
 - Price (Log Transformed)
- **Filter latitude & longitude**

2

DATA ENGINEERING

- **Missing Value Imputation**
 - Random Sampling Imputation
- **Negative Values & Outlier Removal**
 - Discard impossible values
 - Keep the middle 99% data by trimming the top and bottom 0.05%

3

FINAL FEATURES

- **Location**
 - Latitude, Longitude
- **Time**
 - DaysOnMarket, YearBuilt
- **Property**
 - BathroomsTotalInteger, LivingArea, FireplaceYN, YearBuilt, ParkingTotal, BedroomsTotal, PoolPrivateYN, LotSizeAcres, Stories

FINAL MODEL: XGBoost

1 HYPERPARAMETERS

colsample_bytree: 0.8,
learning_rate: 0.05,
max_depth: 7,
n_estimators: 1300,
reg_lambda: 0,
subsample: 0.8,
subsample_freq: 5

2 PERFORMANCE

(on middle 99% test dataset)

R-squared: 0.9

MAPE: 11.23%

MdAPE: 7.76%

Price Bands	MdAPE(%)
<\$500K	7.55
\$500K-\$1M	6.61
\$1M-\$2M	8.9
\$2M-\$5M	11.61

FINAL MODEL: LightGBM

1

MODEL CONFIGURATION

- **Algorithm**
 - LightGBM
- **Split**
 - Jan-Aug (Train)
 - Sep-Oct (Test)
- **Target**
 - Price (Log Transformed)

2

DATA ENGINEERING

- **Missing Value Imputation**
 - Median for numerical columns
 - Mode for categorical columns
- **Negative Values & Outlier Removal**
 - Discard values below 0
 - IQR method

3

FINAL FEATURES

- **Location**
 - Latitude, Longitude
- **Time**
 - DaysOnMarket, YearBuilt
- **Property**
 - BathroomsTotalInteger, LivingArea, FireplaceYN, YearBuilt, ParkingTotal, BedroomsTotal, PoolPrivateYN, LotSizeAcres, Stories

FINAL MODEL: LightGBM

1 HYPERPARAMETERS

colsample_bytree: 1.0,
learning_rate: 0.1,
max_depth: 8,
n_estimators: 1200,
reg_lambda: 0.1,
subsample: 1.0,
subsample_freq: 5

2 PERFORMANCE

(on middle 99% dataset)

R-squared: 0.87

MAPE: 11.08%

MdAPE: 7.71%

Price Bands	MdAPE(%)
<\$500K	7.8
\$500K-\$1M	6.6
\$1M-\$2M	9.08
\$2M-\$5M	12.04

Model Comparison

Model	R-squared	MdAPE(%)	MAPE(%)
XGBoost	0.90	7.76	11.23
LightGBM	0.87	7.71	11.08


$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{H_i - P_i}{H_i} \right|$$

Where:

P_i = predicted house price for property $_i$

H_i = actual house price for property $_i$

$$\bar{H} = \frac{1}{n} \sum_i^n H_i$$

n = number of properties.

Why We Do Not Rely on MAPE

- **House Price Effects on MAPE:**
 - High price homes introduce small errors, pushing MAPE down (Underpenalizing)
 - Low priced homes introduce large errors, pushing MAPE up (Overpenalizing)
- **California housing data is right-skewed:**
 - Many homes in <\$500K and \$500-1M
 - Few homes in higher price categories
- **Result:**
 - Right-skewed data causes an overestimation of the true MAPE for both models
 - MAPE results are drive by the house price distribution, not the model architectures
 - LightGBM is better, but other performance metrics are more meaningful


$$\text{MdAPE} = 100 \cdot \text{median} \left(\left| \frac{H_i - P_i}{H_i} \right| \right)$$

MdAPE is Preferred for % Error Estimation

- **Why:**

- Gives estimate of the typical percent error
- Uses median percentage error (Robust to extreme values)
 - Differences reflect model architecture and specifications

- **Differences in Model Behavior:**

- XGBoost: Slower learning rate (0.05), more estimators (1300), regularization via subsampling
- LightGBM: Faster learning rate (0.10), less estimators (1200), regularization via l-1

- **Interpretation:**

- XGBoost: Slightly flexible → larger adjustments → 7.76% MdAPE
- LightGBM: Slightly more conservative → smaller adjustments → 7.71% MdAPE
- LightGBM is slightly better but difference is effectively equivalent



$$R^2 = 1 - \frac{\sum_{i=1}^n (H_i - P_i)^2}{\sum_{i=1}^n (H_i - \bar{H})^2}$$

R-squared Captures Market Structure

- **Why:**

- Measures how much of the closing price variation is captured by the model
- Higher R-squared means the model understands more of the underlying California housing market patterns

- **Differences in Model Behavior:**

- XGBoost: Slower learning rate (0.05), more estimators (1300), regularization via subsampling
- LightGBM: Faster learning rate (0.10), less estimators (1200), regularization via l-1

- **Interpretation:**

- XGBoost: Learns smooth structure → Captures 90% of variation → More generalizable
- LightGBM: Learns less smooth structure → Captures 87% of variation → Less generalizable
- XGBoost is slightly better

Summary

- **XGBoost**

- Data: Used random sampling imputation for missing values
 - Prevents shrinkage of the test distribution
- Specifications: Lower learning rate, more iterations, feature/row subsampling
- Results: Reduced overfitting → Deeper learning of structure → Improved generalizability

- **LightGBM**

- Data: Used mode/median imputation for missing values
- Specifications: Faster learning rate, full subsampling, leaf-wise splitting
- Results: Computationally cheaper → More aggressive → Lower Percentage Errors

Future Work: Determining the Best Model

- Both models are quite strong
- **Conduct additional tests to determine the best:**
 - K-fold CV for more robust performance metrics
 - Assess consistency across price bands (Get more high-priced homes data)
 - Perform temporal validation
 - Use explainability methods (**SH**apley **A**dditive ex**P**lanations)
- Select model based on **stability, robustness, and interpretability**

Future Work: Model Deployment

- Existing Streamlit interface deployed on a test model

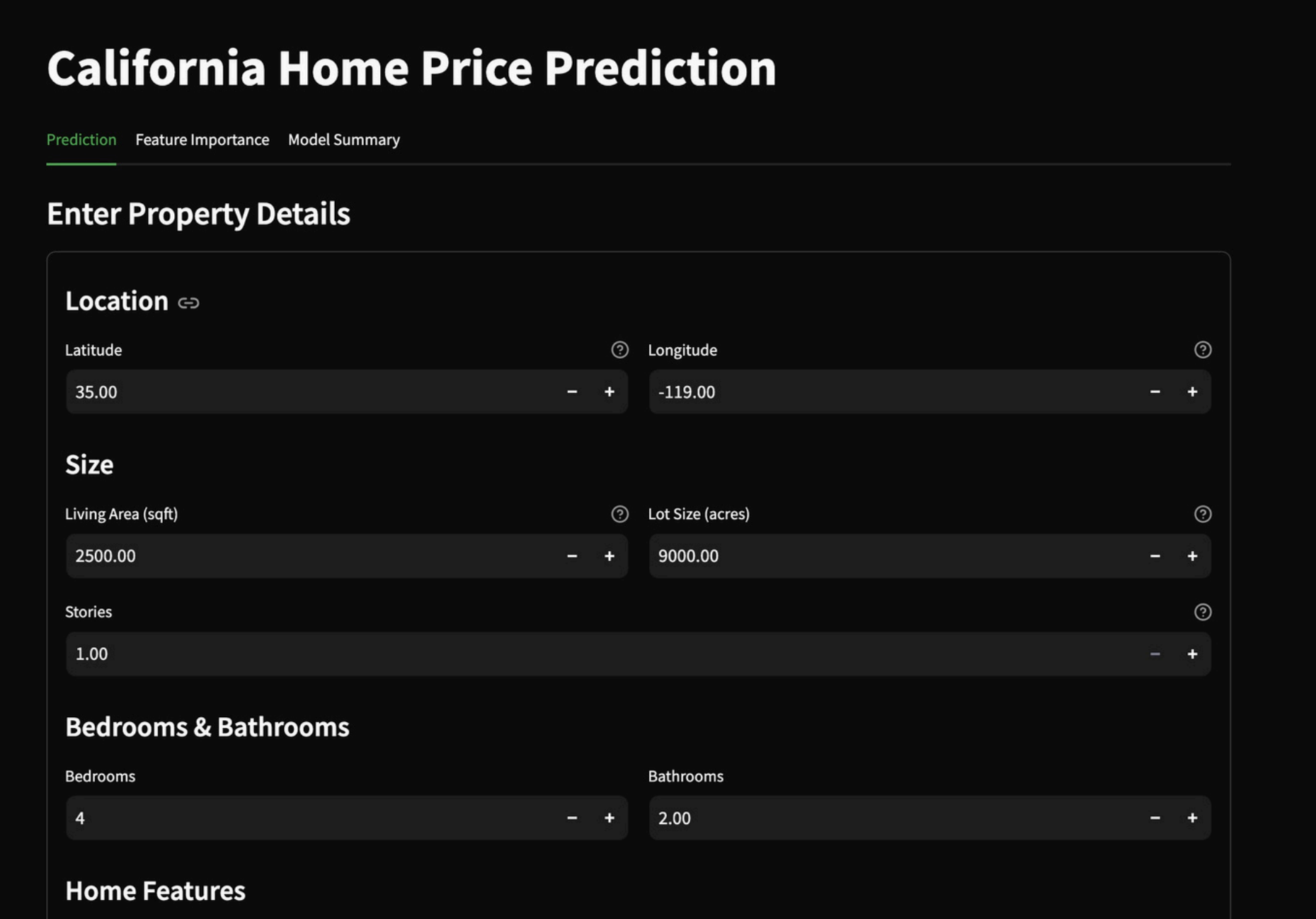
- Includes:

- User input validation
- Feature importance visualization
- Experimental XGBoost model

- [Link](#)

- Next:

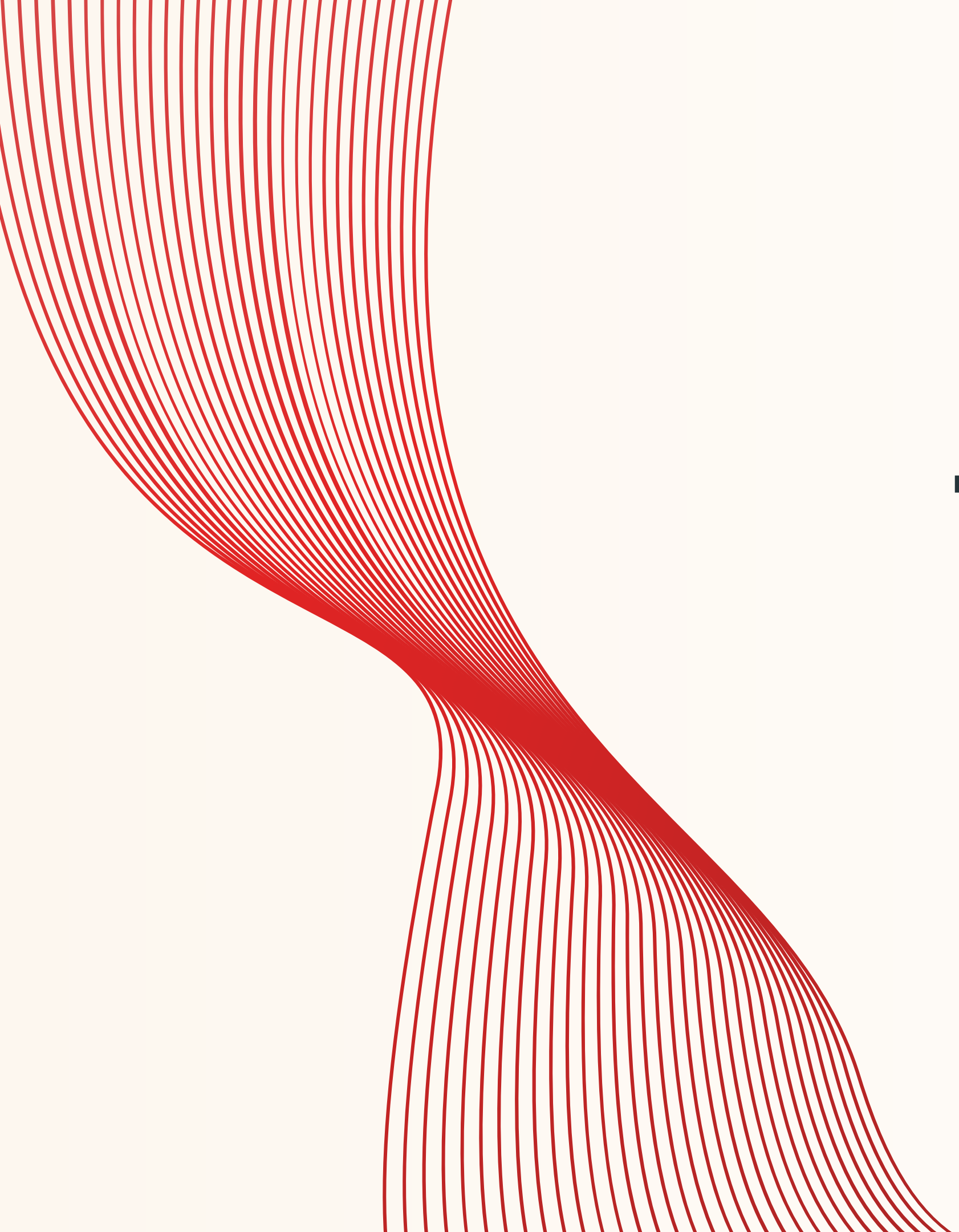
- Integrate full production model
- Enable live predictions
- Enable SHAP explanations



The screenshot displays a Streamlit web application titled "California Home Price Prediction". At the top, there are three tabs: "Prediction" (highlighted in green), "Feature Importance", and "Model Summary". Below the tabs is a section titled "Enter Property Details". This section contains several input fields with numerical values and range controls (minus and plus buttons):

- Location** (indicated by a double-headed arrow icon):
 - Latitude: 35.00
 - Longitude: -119.00
- Size**:
 - Living Area (sqft): 2500.00
 - Lot Size (acres): 9000.00
- Stories**: 1.00
- Bedrooms & Bathrooms**:
 - Bedrooms: 4
 - Bathrooms: 2.00

At the bottom of the visible section is a header for "Home Features".



THANK YOU

